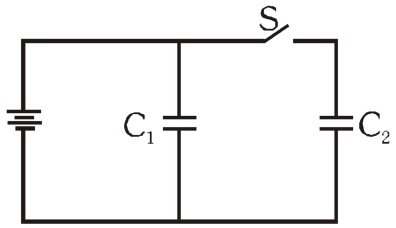
PART-1 : PHYSICS

SECTION-I (i)

1) Two capacitors of equal capacitance (C_1=C_2) are shown in the figure. Initially, while the switch S is open, one of the capacitors is uncharged and the other carries charge Q_0. The energy stored in the charged capacitor is U_0. Sometime after the switch is closed, the capacitors C_1 and C_2 carry charges Q_1 and Q_2, respectively; the voltage across the capacitors are V_1 and V_2; and the energies stored in the capacitors are U_1 and U_2. Which of the following statements is incorrect ?



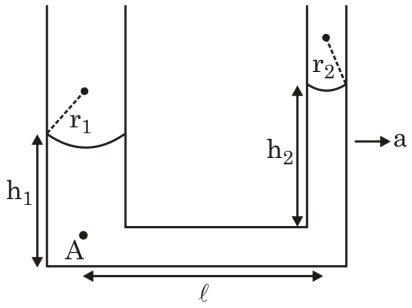
(A) $Q^0 = \frac{1}{2}(Q_1 + Q_2)$

(B) Q_1=Q_2

(C) V_1=V_2

(D) $U_0 = U_1 + U_2$

2) A vertical communicating tube contains a liquid of density ρ and surface tension T. If it moves with a horizontal acceleration “a”, pressure at ‘A’ is equal to :-



(A) $\rho g h_2 + P_0 + \frac{2T}{r_2} + \rho a l$

(B) $\rho g h_2 - \rho a l + P_0 + \frac{2T}{r_2}$

(C) $\rho g h_1 + P_0 - \frac{2T}{r_1}$

(D) $\rho g h_1 - P_0 - \frac{2T}{r_1}$

3) The fundamental frequency of a closed organ pipe is same as the first overtone frequency of an open pipe. If the length of open pipe is 50 cm, the length of closed pipe is

(A) 25 cm

(B) 12.5 cm

(C) 100 cm

(D) 200 cm

4) The electric and the magnetic field, associated with an e.m. wave, propagation along the +z-axis, can be represented by–

(A) $\left[\vec{E} = E_0\hat{i}, \vec{B} = B_0\hat{j}\right]$

(B) $\left[\vec{E} = E_0\hat{k}, \vec{B} = B_0\hat{i}\right]$

(C) $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{i}\right]$

(D) $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{k}\right]$

SECTION-I (ii)

1) In Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes as 9. This implies :

(A) the intensities at the screen due to the two slits are 5 and 4 units

(B) the intensities at the screen due to the two slits are 4 and 1 units

(C) the amplitude ratio is 3

(D) the amplitude ratio is 2

2) A parallel plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges +Q and –Q respectively. A third plate C, identical to A and B, with charge +Q, is now introduced midway between A and B, parallel to them. Which of the following is/are correct?

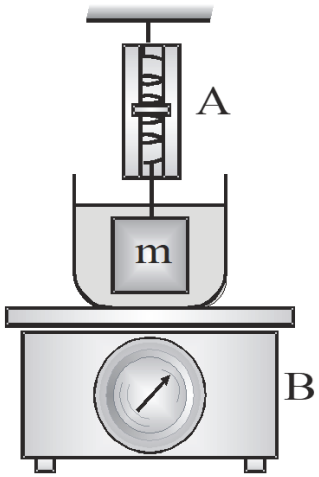
(A) Charge on the inner face of B is now $-\frac{3Q}{2}$

(B) There is no change in the potential difference between A and B

(C) Potential difference between A and C is one–third of the potential difference between B and C

(D) Charge on the inner face of A is now $\frac{Q}{2}$

3) The spring balance A read 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in figure. In this situation :–



(A) The balance A will read more than 2 kg

(B) The balance B will read more than 5 kg

(C) The balance A will read less than 2 kg and B will read more than 5 kg

(D) The balance A and B will read 2 kg and 5 kg respectively.

SECTION-II (i)

Common Content for Question No. 1 to 2

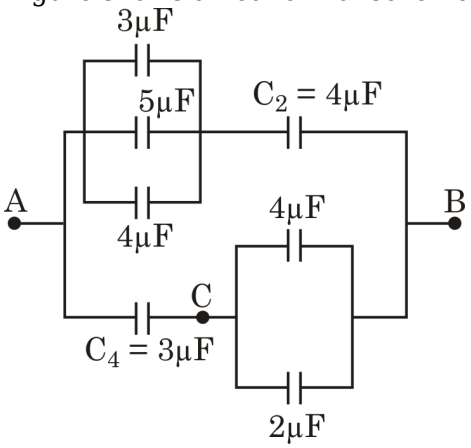
A thin paper of thickness 0.02 mm having a refractive index 1.45 is pasted across one of the slits in a Young's double slit experiment. The paper transmits $\frac{4}{9}$ of the light energy falling on it.

1) The ratio of maximum to minimum intensity in the fringe pattern is :-

2) How many fringes will cross through the centre if an identical paper piece is pasted on the other slit also? The wavelength of the light used is 6000Å.

Common Content for Question No. 3 to 4

Figure shows a network of seven capacitors. The charge on the 5 μF capacitor is 10 μC.



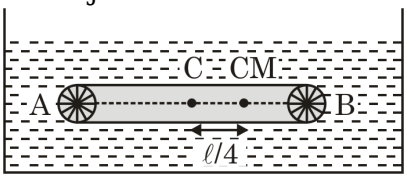
3) The potential difference (in V) between points A and B is :-

4) The equivalent capacitance between points A and B is (in μF) :-

SECTION-II (ii)

1) One slit of a Young's experiment is covered by a glass plate ($\mu_1 = 1.4$) and the other by another glass plate ($\mu_2 = 1.7$) of the same thickness. The point of central maxima on the screen, before the plates were introduced is now occupide by the third bright fringe. Find the thickness of the plates (in μm), when the wavelength of light used is 4000 Å.

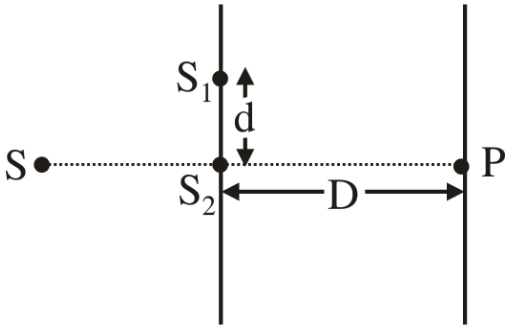
2) A non uniform cylinder of mass m, length and radius r is having its centre of mass at a distance $\ell/4$ from the centre and lying on the axis of the cylinder. The cylinder is kept in a liquid of uniform density ρ. The moment of inertia of the rod about the centre of mass is I. The angular acceleration of point A relative to point B just after the rod is released from the position shown in figure is $\frac{\pi\rho g\ell^2 r^2}{nI}$.



Find the value of n.

3) A block of mass 0.5 kg is attached to one end of a cord of mass 3.2×10^{-5} kg; the other end of the cord is attached to a fixed point. The block rotates with constant angular speed in a circle on a horizontal, frictionless table. Through what angle does the block rotate in the time interval required for a transverse wave to travel along the string from the center of the circle to the block? Write answer in form $x \times 10^{-3}$ rad. The nearest integer value of x is :-

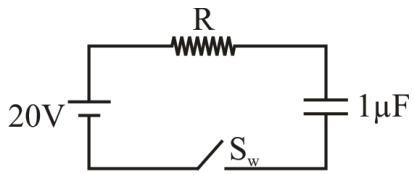
4) In a YDSE experiment two slits S_1 and S_2 have separation of d = 2 mm. The distance of the screen is D = 8/5 m. Source S starts moving from a very large distance towards S_2 perpendicular to S_1S_2 as shown in figure. The wavelength of monochromatic light is 500 nm. The number of maximas observed on the screen at point P as the source moves towards S_2 is 3995 + n. Find the value of n.



5) Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0 m. The separation between the positions of the first minima of diffraction pattern obtained in the two cases is _____ $\times 10^{-5}$ m.

6) Figure shows an uncharged capacitor connected in series with a resistor R and ideal battery. At t = 0 switch S_W is closed and it is observed that when charge on capacitor is 75% of maximum value

then current in the circuit is 1A, then value of R in Ω is :-



PART-2 : CHEMISTRY

SECTION-I (i)

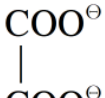
1) In which of the following molecules/ions all the bonds are not of equal length?

(A) SiF_4

(B) SF_4

(C) XeF_4

(D) BH_4^-

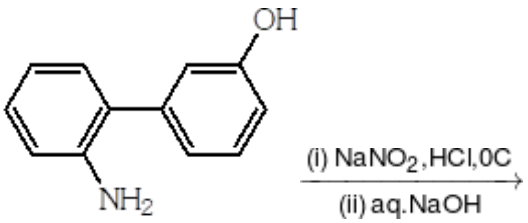
2) Given at infinite dilution the ionic conductance of  , $K^{\oplus}$, and $Na^{\oplus}$ are 74, 50, and 73 $cm^2\ ohm^{-1}\ eq^{-1}$, respectively. The equivalent conductance at infinite dilution of the salt  is :

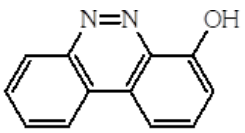
(A) 197 $cm^2\ ohm^{-1}\ eq^{-1}$

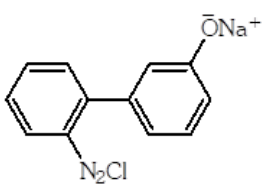
(B) 172 $cm^2\ ohm^{-1}\ eq^{-1}$

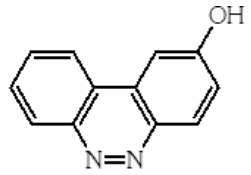
(C) 135.5 $cm^2\ ohm^{-1}\ eq^{-1}$

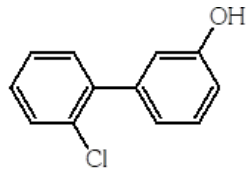
(D) 160.5 $cm^2\ ohm^{-1}\ eq^{-1}$

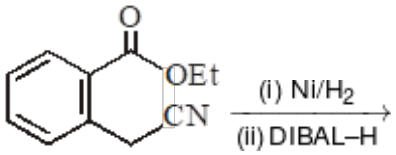
3) The major product of the following reaction is 

(A) 

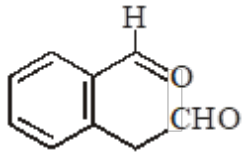
(B) 

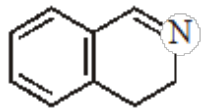
(C) 

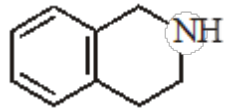
(D) 

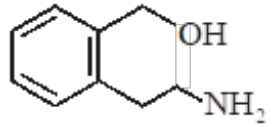
4) The major product 'R' of the following reaction is 

R

(A) 

(B) 

(C) 

(D) 

SECTION-I (ii)

1) Cu^{2+} ions give white precipitate with :

(A) Excess KI solution
(B) Potassium thiocyanate solution
(C) Excess KCN solution
(D) Potassium hydroxide solution

2) Which of the following statement(s) is/are correct ?
In the reaction $xCu_3P + yCr_2O_7^{2-} \rightarrow Cu^{2+} + H_3PO_4 + Cr^{3+}$

(A) Cu in Cu_3P is oxidised to Cu^{2+} whereas P in Cu_3P is also oxidised to PO_4^{3-}.
(B) Cu in Cu_3P is oxidised to Cu^{2+} whereas P in Cu_3P is reduced to H_3PO_4.
(C) In the conversion of one molecule of Cu_3P to Cu^{2+} and H_3PO_4, 11 electrons are involved.
(D) The value of x is 6, if reaction is balanced by minimum integral coefficients.

3) Which of the following is/are correct statement(s) ?

(A) If a solution is prepared by dissolving excess of $AgNO_3$ solution in KI solution, the charge likely to develop on colloidal particle is positive.

(B) Ultracentrifugation process is used for preparation of lyophobic sol.

(C) Gold number is the index for the extent of gold-plating.

(D) Lyophobic sols are usually less solvated than lyophilic sol.

SECTION-II (i)

Common Content for Question No. 1 to 2

An aqueous solution of strong acid containing 0.01 M Fe^{3+} and 0.01 M Fe^{2+} was titrated with a concentrated solution of NaOH at 30°C, so that changes in volumes were negligible. Assuming that the new species formed during titration are $Fe(OH)_3$ and $Fe(OH)_2$ only. $\left[\frac{RT(2.303)}{F} = 0.06\right]$

Given : $E^{\ominus}_{Fe^{3+}|\,Fe^{2+}} = 0.80\,V$, $K_{sp\ Fe(OH)_3} = 10^{-37}$, and $K_{sp\ Fe(OH)_2} = 10^{-19}$

1) Calculate the redox potential in millivolts of Fe^{3+} | Fe^{2+} electrode at pH = 2

2) The redox potential of Fe^{3+} | Fe^{2+} electrode in millivolts at pH = 4 is :

Common Content for Question No. 3 to 4

An organic acid P ($C_{11}H_{12}O_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, P gives an apliphatic ketone as one of the products. P undergoes the following reaction sequences to furnish R via Q. The compound P also undergoes another set of reactions to produce S.

$$S \xleftarrow[(iv)\,CHCl_3, KOH, \Delta]{\substack{(i)\,H_2/Ni \\ (ii)\,NH_3/\Delta \\ (iii)\,Br_2/NaOH \\ (v)\,LiAlH_4/Et_2O}} P \xrightarrow[(iii)\,MeMgBr, CdCl_2]{\substack{(i)\,H_2/Ni \\ (ii)\,SOCl_2 \\ (iv)\,NaBH_4}} Q \xrightarrow[(iii)\,CO_2\,(dry\ ice)]{\substack{(i)\,HCl \\ (ii)\,Mg/Et_2O \\ (iv)\,H_3O^+}} R$$

3) The number of π-electrons in compound R.

4) The DBE of the compound S is :

SECTION-II (ii)

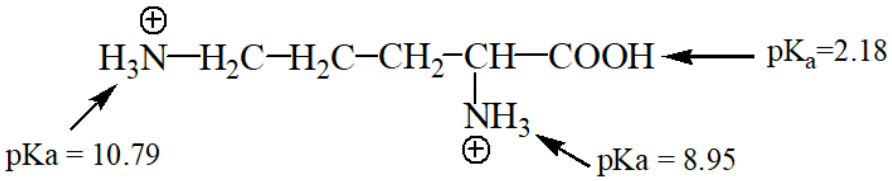
1) The choice of a reducing agent in a particular case depends on thermodynamic factor.
Given ΔG^{o}_{f} Al_2O_3 = –900 kJ mol^{-1} and ΔG^{o}_{f}, Cr_2O_3 = –500 kJ mol^{-1}
Among Al and Cr, find the amount of metal (in Kg) that can reduce 3000 moles of oxides of other metal, in standard conditions. (Al = 27, Cr = 52)

2) In the complex $[Fe(Cl)(CN)_4(O_2)]^{4-}$ the electronic configuration is :

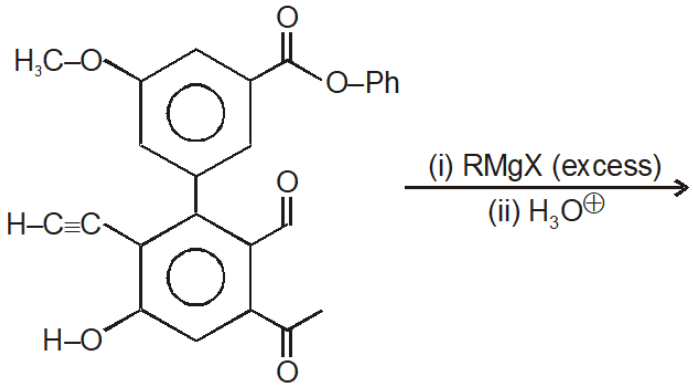
$t_{2g}^{2,2,2}\ e_g^{0,0}$. Find spin only magnetic moment of the complex in B.M. Report your answer as nearest integer.

3) Hydrogen sulphide reacts with iodic acid in an aqueous solution liberating iodine. In an experiment 20 cc of a solution of iodic acid was allowed to react completely with an excess of H_2S. The excess of H_2S and iodine formed were removed by heating the solution. The H_2SO_4 present in the solution required 35.5 cc of 0.16 N NaOH solution for complete neutralisation. Calculate the amount of HIO_3 (in gm per litre) of the original solution. (I = 127)
(Answer in nearest integer)

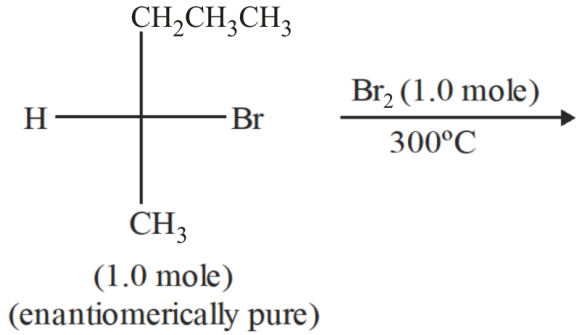
4) If pKa values for protonated Lysine molecule are pKa_1 = 2.18 ; pKa_2 = 8.95 ; pKa_3 = 10.79 respectively then , Calculate value of (x – 1) where x is isoelectronic point of lysine.



5) Number of R-MgX (grignard reagent) consumed are



6) In the following monobromination reaction, the number of possible chiral products is



PART-3 : MATHEMATICS

SECTION-I (i)

1) The area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = 4$, the x-axis and the line $y = x/\sqrt{3}$ is :

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) 1

2) Let $S(x) = \int \frac{dx}{e^x + 8e^{-x} + 4e^{-3x}}$, $R(x) = \int \frac{dx}{e^{3x} + 8e^x + 4e^{-x}}$ and $M(x) = S(x) - 2R(x)$.
If $M(x) = \frac{1}{2}\tan^{-1}\left(\frac{f(x)}{2}\right) + c$, (where c is arbitrary constant) and $M(0) = \frac{1}{2}\tan^{-1}\left(\frac{3}{2}\right)$, then $f(\log_e 2)$ is equal to

(A) 1

(B) 3

(C) 5

(D) 7

3) Let f be a differentiable function satisfies $\int_0^x (x - t + 1)\, f(t)dt - \int_0^x \left(4t^3 + 2t\right) dt = 0$, $\forall\, x > 0$ then the value of $\frac{12}{\pi}\int_0^1 \frac{e^x dx}{\frac{d}{dx}(f(x).e^x) + 10e^x} =$

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{8}$

4) The solution of the differential equation $e^{2y}(1 + \square nx)dx + cosecy(2 + coty)dy = 0$ satisfying $y(1) = \frac{\pi}{2}$ is

(A) $x\,\ell n\, x + e^{-\pi} = \frac{e^{-2y}}{\sin y}$

(B) $2x\,\ell nx + e^{-\pi} = \frac{e^{-2y}}{\sin y}$

(C) $\frac{x}{2}\ell n\, x + e^{-\pi} = \frac{e^{-2y}}{\sin y}$

(D) $\frac{3\ell n\, x}{x} + e^{-\pi} = \frac{e^{-2y}}{\sin y}$

SECTION-I (ii)

1) Let $I_1 = \int_0^{\ln 10^4} \left(e^{2x} - e^{x}[e^{x}]\right) dx$ and $I_2 = \int_0^{9\times 10^6} \left(1 - \frac{[\sqrt{x}]}{\sqrt{x}}\right) dx$. Then which of the following is/are correct.

where [•] denotes greatest integer function.

(A) $I_1 > I_2$

(B) $\frac{I_1}{I_2} = \frac{3}{5}$

(C) $\left[\frac{I_1 - I_2}{1000}\right] = 1$

(D) $\frac{I_1}{I_2} = \frac{5}{3}$

2) Let S_k be the area bounded by the curve $y = x^2 (1 - x)^k$ and the lines $x = 0$, $y = 0$ and $x = 1$. Then which of the following is/are correct.

(A) $\sum_{k=1}^{100} S_k < \frac{1}{6}$

(B) If $\lim_{n\to\infty} \sum_{k=1}^{n} S_k = \frac{p}{q}$, then $p + q = 7$ where p and q are coprime numbers

(C) $S_{99} = \frac{1}{100 \times 101 \times 102}$

(D) $\sum_{k=1}^{999} S_k > \frac{1}{6}$

3) Equation of curve passing through (3,4) and satisfying the differential equation $y\left(\frac{dy}{dx}\right)^2 + (x - y)\frac{dy}{dx} - x = 0$ can be

(A) $x^2 + y^2 = 25$

(B) $x^2 - y^2 + 7 = 0$

(C) $xy = 12$

(D) $x - y + 1 = 0$

SECTION-II (i)

Common Content for Question No. 1 to 2

Consider a twice differentiable function f(x) satisfying $f(x) + f''(x) - 2f'(x) = 0$ where $f(0) = 0$ and

f(1) = 3e.

1) The number of zeroes of the equation f(x) = 0 is equal to

2) $\int_0^1 \left(-\frac{e^x x^3}{2} + f(x)\right) dx$ is equal to λe then λ is(where 'e' is Napier's Constant)

Common Content for Question No. 3 to 4

Let $f(x) = \begin{cases} x^3 & ; \quad x \geqslant 0 \\ x^2 & ; \quad x < 0 \end{cases}$ and $g(x) = \begin{cases} x^2 & ; \quad x > 0 \\ x^3 & ; \quad x \leqslant 0 \end{cases}$.

3) Area bounded by y = f(x) and y = g(x) is λ, then the value of 144λ is -

4) Area bounded by f(x) and line y = 1 is μ, then the value of 12μ is -

SECTION-II (ii)

1) Let function $f : R \to R$ defined by $f(x) = 2x^3 + bx^2 + 2x + \int_{-x}^{4} (2\sin^2 t + 3\cos^2 t)dt, \forall x \in R$. If f(x) is strictly increasing function for ∀ x ∈ R, then number of integral values of b is

2) If $k = \int_{-2013}^{2013} \frac{dx}{1 + x^{2011} + \sqrt{1 + x^{4022}}}$ then the sum of digit of value k is

3) If the possible solution of the differential equation $y(y^2 - 2x^2)dx + x(2y^2 - x^2)dy = 0$ is of the form f(x).g(y).(g(y) – f(x)) = C where f(x), g(y) are monic polynomials. then find value of g(3) + f(2).

4) Let f : [–3, 1] → R be given as

$$f(x) = \begin{cases} \min\left\{(x+6), x^2\right\}, & -3 \leqslant x \leqslant 0 \\ \max\left\{\sqrt{x}, x^2\right\}, & 0 \leqslant x \leqslant 1. \end{cases}$$

If the area bounded by y = f(x) and x-axis is A, then the value of 6A is equal to

5) The value of the integral $\int_0^{\frac{\pi}{6}} \frac{\left(2\sqrt{2}\cos 15^\circ\right)\cos x \, dx}{\left((\sin x + 1)^{1010}(1 - \sin x)^{3030}\right)^{\frac{1}{2020}}}$ is

6) Let $p = \lim_{n\to\infty} \frac{1}{n}((1+n)(2+n)\ldots(2n))^{\frac{1}{n}}$. The area bounded by line x = ep and y = ep and parabola $y^2 = 16x$ in the I^{st} quadrant is A, then value of [A] is
(Where [.] denotes greatest integer function)

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	D	C	B	A

SECTION-I (ii)

Q.	5	6	7
A.	B,D	A,B,C,D	B,C

SECTION-II (i)

Q.	8	9	10	11
A.	25.00	15.00	8.00	5.00

SECTION-II (ii)

Q.	12	13	14	15	16	17
A.	4	4	8	5	2	5

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	18	19	20	21
A.	B	C	C	B

SECTION-I (ii)

Q.	22	23	24
A.	A,B	A,C,D	A,D

SECTION-II (i)

Q.	25	26	27	28
A.	800.00	500.00	8.00	4.00

SECTION-II (ii)

Q.	29	30	31	32	33	34
A.	162	2	40	9	6	5

PART-3 : MATHEMATICS

SECTION-I (i)

Q.	35	36	37	38
A.	**B**	**B**	**C**	**A**

SECTION-I (ii)

Q.	39	40	41
A.	**A,C**	**A,B**	**A,D**

SECTION-II (i)

Q.	42	43	44	45
A.	**1.00**	**1.00**	**12.00**	**17.00**

SECTION-II (ii)

Q.	46	47	48	49	50	51
A.	**9**	**6**	**13**	**41**	**2**	**6**

SOLUTIONS

PART-1 : PHYSICS

1)

Both capacitor are in parallel so $v_1 = v_2$ and both have capacitance equal.

So $Q_1 = C_1V_1$; $Q_2 = C_2V_2$

$Q_1 = Q_2$

Initially $Q_0 = C_1V = CV$

Finally $Q_1 = C_1V$ and $Q_2 = C_2V$ $[C_1 = C_2]$

$$Q_0 = \frac{1}{2}(Q_1 + Q_2)$$

$$U_0 = \frac{1}{2}CV^2 \quad \text{....(1)}$$

$$U_1 = \frac{1}{2}CV^2 \quad \text{....(2)}$$

$$U_2 = \frac{1}{2}CV^2$$

$$U_0 \neq U_1 + U_2 \left[\frac{1}{2}CV^2 \neq CV^2\right]$$

2) $P_0 - P_1 = \frac{2T}{r_1}$

$P_A = P_1 = \rho g h_1$

5) $$\frac{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = 9 \Rightarrow \frac{I_1}{I_2} = 4 \text{ and } I \propto (\text{Amplitude})^2$$

$\sqrt{I_1} + \sqrt{I_2} = (3)\left(\sqrt{I_1} - \sqrt{I_2}\right)$

$4\sqrt{I_2} = 2\sqrt{I_1}$

$I_1 = 4I_2$

6) $C_{initial} = \frac{a\in_0}{2x}$

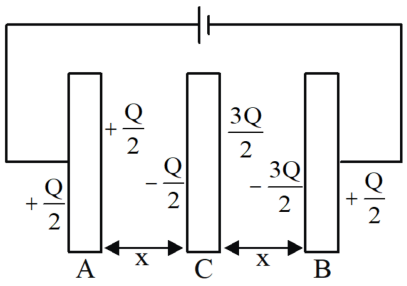
After inserting plate C

$$C_{eq} = \frac{\frac{a\varepsilon_0}{x} \times \frac{a\varepsilon_0}{x}}{\frac{2a\varepsilon_0}{x}}$$

$$C_{eq} = \frac{a\varepsilon_0}{2x}$$

Hence, Battery will not given any extra charge became capacitance is same before and after

inserting plate C.



7) Reading of spring balance A is = $(mg - F_b)$
$(F_s)_A < 2$ kg –wt
Reading of spring balance B = $(m_2g + F_b)$
$(F_s)_B > 5$ kg – wt

8) $I_1 = I$ and $I_2 = \frac{4I}{9}$.

$$\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2} = \frac{9}{4}; \quad \frac{A_1}{A_2} = \frac{3}{2}$$

$$\text{Thus} \quad \frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(3+2)^2}{(3-2)^2} = 25$$

9)

Displacement of fringes,

$$\Delta = \frac{D(\mu - 1)t}{d}, \text{ and}$$

fringe width, $\beta = \frac{D\lambda}{d}$

$$\therefore \quad n = \frac{\Delta}{\beta} = \frac{D(\mu - 1)t/d}{(D\lambda/d)}$$

$$= \frac{(\mu - 1)t}{\lambda}$$

$$= \frac{(1.45 - 1) \times 0.02 \times 10^{-3}}{6000 \times 10^{-10}} = 15.$$

10) Given, charge on 5 μF capacitor = 10 μC = 10×10^{-6}C.

∴ Potential difference across 5 μF capacitor = $\frac{10 \times 10^{-6}}{5 \times 10^{-6}}$ = 2V

As the 3 μF, 4 μF and 5 μF capacitors are joined in parallel, the potential difference across each is the same = 2 V. Therefore,

Charge on 3 μF capacitor = $3 \times 10^{-6} \times 2 = 6 \times 10^{-6} = 6$ μC

Charge on 4 μF capacitor = $4 \times 10^{-6} \times 2 = 8 \times 10^{-6}$C = 8 μC

∴ Total charge flowing through C_1 and C_2 = 10 μC + 6 μC + 8 μC = 24 μC.

∴ Potential difference across C_2 = $\frac{24\mu C}{4\mu F}$ = 6 V

∴ Total potential difference across AB = 2 V + 6 V = 8V

11) The equivalent capacitance C_1 of the parallel combination of 3 μF, 5 μF and 4 μF capacitors is

$C_1 = 3 + 5 + 4 = 12$ μF

The equivalent capacitance C_3 of the parallel combination of 4 μF and 2 μF capacitors is

$C_3 = 4 + 2 = 6$ μF

Now C_1 and C_2 are in series. Their equivalent capacitance C' is given by

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{12} + \frac{1}{4} = \frac{1}{3}$$

or C = 3 μF. The equivalent capacitance C" of the series combination of C_3 and C_4 is given by

$$\frac{1}{C''} = \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

or C" = 2 μF. The equivalent capacitance between A and B = capacitance of a parallel combination of C' and C" = C' + C" = 3 + 2 = 5 μF.

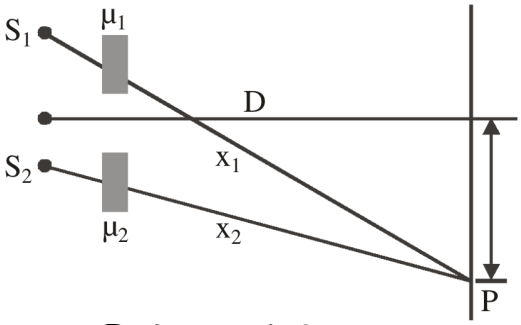
12) Optical path difference

$= [(S_2P - t) + \mu_2 t)] - [(S_1P - t) + \mu_1 t)]$

$$= S_2P - S_1P + (\mu_2 - \mu_1)t = \frac{Y \cdot d}{D} + (\mu_2 - \mu_1)t$$

Let P be the point where optical path difference is zero.

$$\Rightarrow \frac{Y \cdot d}{D} + (\mu_2 - \mu_1) \cdot t = 0$$



$$\Rightarrow Y = \frac{D \cdot (\mu_2 - \mu_1) \cdot t}{d}$$

But Y = 3w (where w is the fringe width 0.

$$Y = 3\frac{D\lambda}{d}$$

$$\Rightarrow 3\frac{D\lambda}{d} = \frac{D}{d}(\mu_2 - \mu_1).t$$

$$t = \frac{3\lambda}{(\mu_2 - \mu_1)} = \frac{3 \times 4000 \times 10^{-10}}{(1.7 - 1.4)} = 4\mu m$$

13) $\left((\pi r^2)L\rho g\right)\frac{L}{4} = I\alpha$

$$\alpha = \frac{\pi r^2 \ell^2 \rho g}{4I}$$

14)

$$\theta = \omega t = \omega\left(\frac{\ell}{v}\right) = \frac{\omega\ell}{\sqrt{\frac{T}{\mu'}}} = \omega\ell\sqrt{\frac{\mu}{M\omega^2\ell}}$$

$$= \sqrt{\frac{m}{M}} = \sqrt{\frac{3.2 \times 10^{-5}}{0.5}}$$

$$= \sqrt{\frac{320}{45}} \times 10^{-2} = \frac{8}{3} \times 10^{-2} = 2.66 \times 10^{-2}$$

15) $S_1P–S_2P = \frac{d^2}{2D} = \frac{2 \times 10^{-3} \times 2 \times 0^{-3}}{2 \times \frac{8}{5}} = \frac{5}{2}\lambda$ $(\lambda = 500nm)$

So when S is at ∞ there is I[st] minima and when S is at S_2 there is last minima because d/l = 4000

So the number of minima's will be 4001 and number of maxima's will be 4000.

16)

a sin θ = λ

$\frac{y}{L} = \theta = \frac{\lambda}{a}$ L = 2m

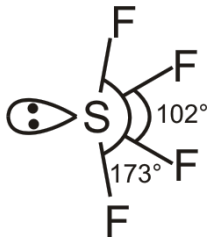
$y = \frac{\lambda L}{a}$

$\Delta y = \frac{\Delta \lambda L}{a}$

$= 2 \times 10^{-5}$ m

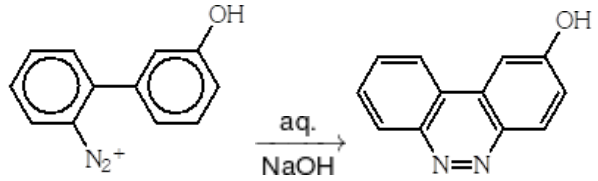
PART-2 : CHEMISTRY

18)

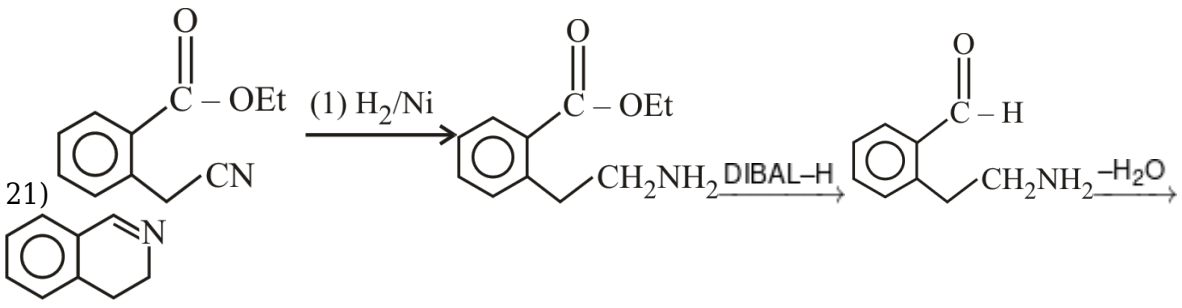


Axial bond length is higher.

20)



21)



22)

(A) Cu_2I_2 (white ppt.)

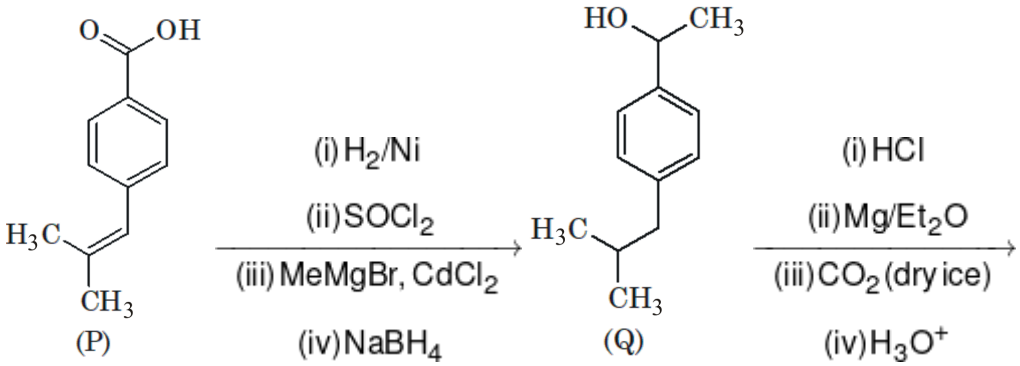
$$2Cu(SCN)_2\downarrow \xrightarrow{\text{decomposes}} \underset{\text{white ppt.}}{2\,CuSCN\downarrow} + (SCN)_2$$

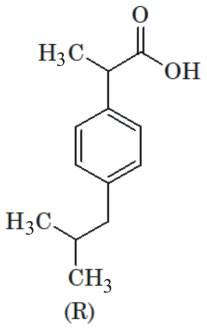
(B) black ppt.

(C) $Cu(CN)_2\downarrow$ yellow ppt.

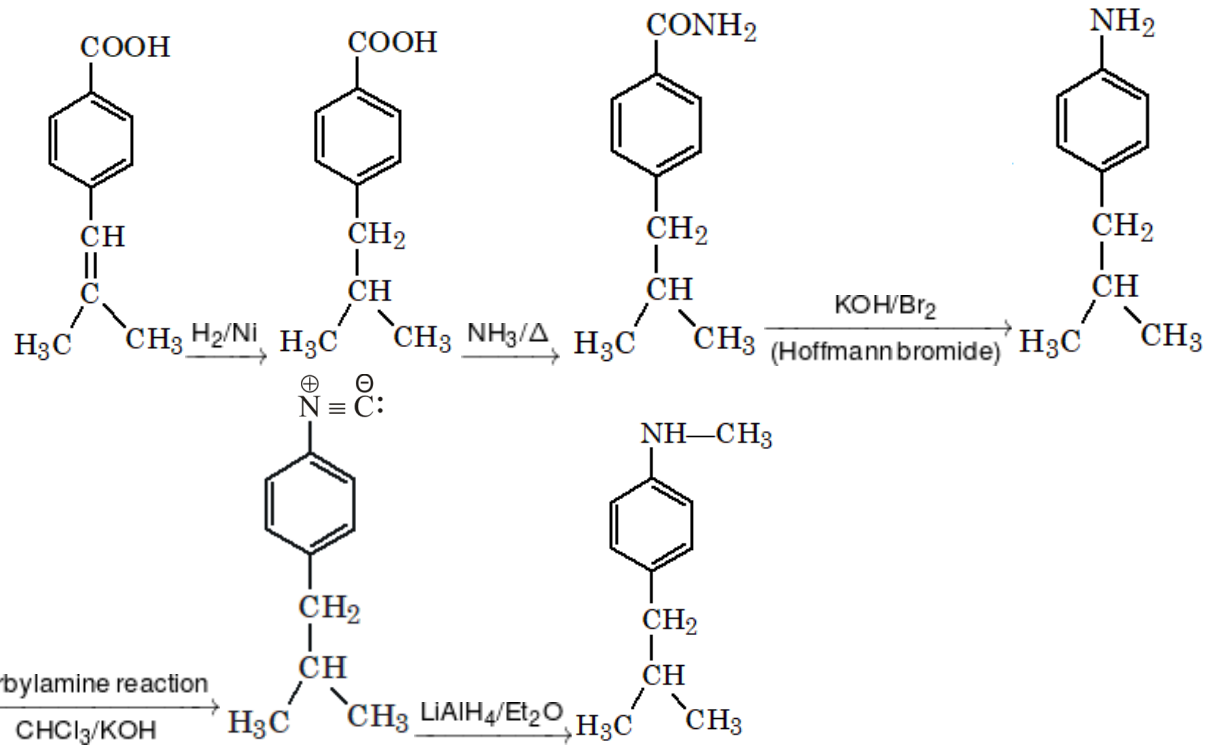
(D) $Cu(OH)_2\downarrow$ blue ppt.

27)





28)



29) Al can reduce Cr_2O_3

Cr_2O_3 + 2Al → Al_2O_3 + 2Cr

3K mole 6K mole

= 6 × 27 Kg = 162 Kg

30) Fe^{2+}

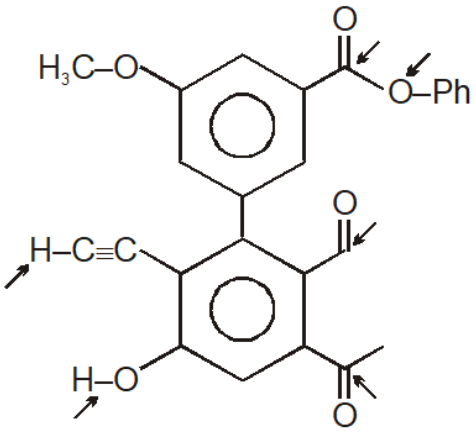
 2 – 1 – 4 + x = –4 ⇒ x = –1

⇒ O_2^{-1} ⇒ Superoxide.

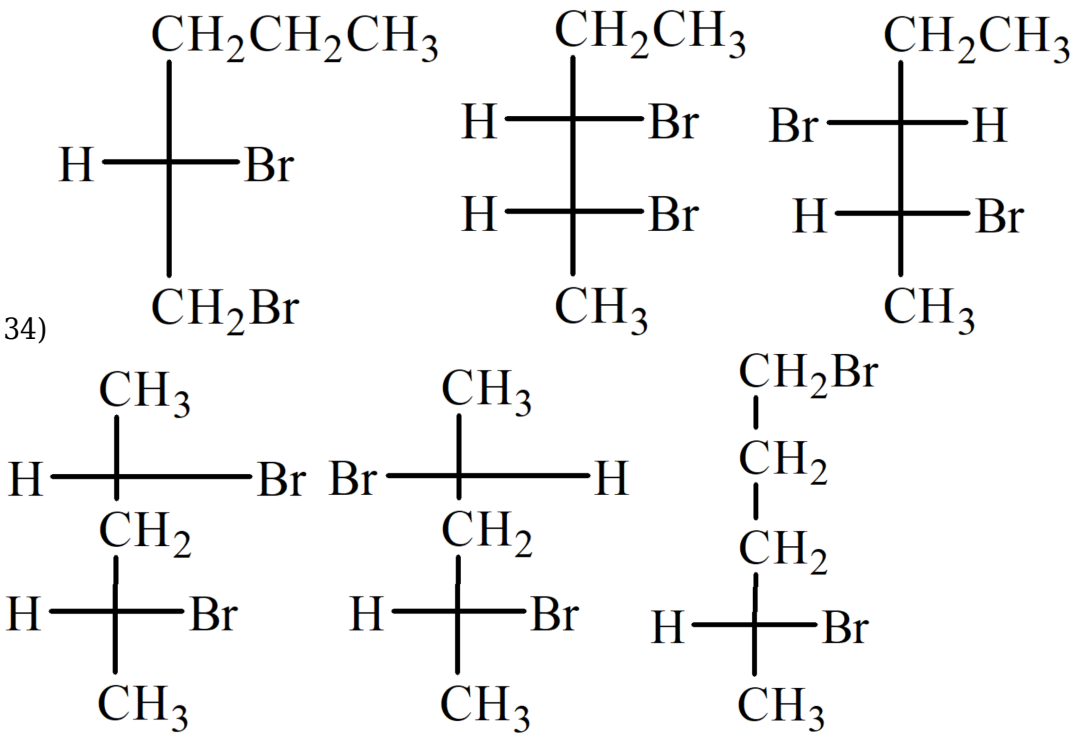
 number of unpaired electron in superoxide = 1

 $\mu = \sqrt{n(n+2)}\ BM = \sqrt{3}\ BM = 1.73\ BM$

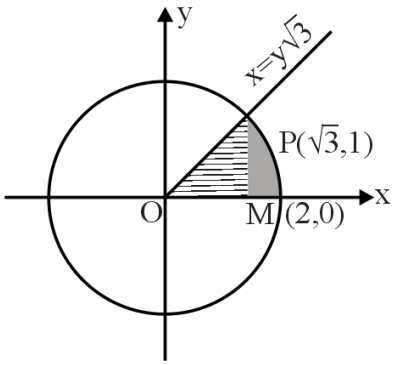
33)



34)



PART-3 : MATHEMATICS



35)

The line $y = \frac{x}{\sqrt{3}}$ meets the circle in the first quadrant at $P(\sqrt{3}, 1)$. Therefore

Area required = $\Delta OPM + \int_{\sqrt{3}}^{2} \sqrt{4-x^2}\,dx$

$= \frac{1}{2} \times \sqrt{3} \times 1 + \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2}\sin^{-1}\frac{x}{2}\right]_{\sqrt{3}}^{2}$

$= \frac{\sqrt{3}}{2} + \left[2\sin^{-1}1 - \frac{\sqrt{3}}{2} - 2\sin^{-1}\frac{\sqrt{3}}{2}\right]$

$= \frac{\sqrt{3}}{2} + \pi - \frac{\sqrt{3}}{2} - 2 \times \frac{\pi}{3}$

36) $M(x) = \int \frac{e^x(e^{2x}-2)}{e^{4x}+8e^{2x}+4}dx$

Put $e^x = t$

$M(x) = \int \frac{t^2-2}{t^4+8t^2+4}dt$

$M(x) = \frac{1}{2}\tan^{-1}\left(\frac{e^x+2e^{-x}}{2}\right) + c$

$M(0) = \frac{1}{2}\tan^{-1}\left(\frac{3}{2}\right)$

$\Rightarrow$ c = 0

$\Rightarrow$ $f(x) = e^x + 2e^{-x}$ $\Rightarrow f(\log_e 2) = 3$

37) $x\int_0^x f(t)dt + \int_0^x (1-t)f(t)dt = x^4 + x^2$

Differentiate with request to x

$x \cdot f(x) + \int_0^x f(t)dt + (1-x)f(x) = 4x^3 + 2x$

Again differentiate with respect to x

$f'(x) + f(x) = 12x^2 + 2$

$\frac{12}{\pi}\int_0^1 \frac{dx}{12(x^2+1)} = \frac{12}{12\times\pi}\tan^{-1}x\Big|_0^1 = \frac{1}{4}$

38) $e^{-2y}\,(2\operatorname{cosec} y + \operatorname{cosec} y \cot y)\,dy = -(1+\log x)\,dx$

Integrating both sides

$-2y = t$

$$\int e^{t}\left(-2\operatorname{cosec}\frac{t}{2} + \operatorname{cosec}\frac{t}{2}\cot\frac{t}{2}\right)dt = 2\int (1+\log x)dx$$

$$\Rightarrow -e^{-2y}\operatorname{cosec}(-y) = x\log x + C$$

$$x = 1,\ y = \frac{\pi}{2} \Rightarrow C = e^{-\pi}$$

$$I_1 = \int_0^{\ln 10^4} e^x\{e^x\}dx$$

39)

$$e^x = t \quad \Rightarrow \quad e^x\,dx = dt$$

$$= \int_1^{10^4} \{t\}\,dt = (10^4 - 1)\int_0^1 t\,dt = \frac{9999}{2}$$

$$I_2 = \int_0^{9\times 10^6} \frac{\{\sqrt{x}\}}{\sqrt{x}}dx$$

$$\sqrt{x} = t \quad \Rightarrow \quad \frac{dx}{\sqrt{x}} = 2dt$$

$$= \int_0^{3000} 2\{t\}\,dt$$

$$= 6000\int_0^1 t\,dt = 3000$$

$$S_k = \int_0^1 x^2(1-x)^k dx = \int_0^1 (1-x)^2 x^k dx$$

40)

$$= \int_0^1 \left(x^k - 2x^{k+1} + x^{k+2}\right)dx$$

$$= \frac{1}{k+1} - \frac{2}{k+2} + \frac{1}{k+3}$$

$$S_k = \left(\frac{1}{k+1} - \frac{1}{k+2}\right) - \left(\frac{1}{k+2} - \frac{1}{k+3}\right)$$

$$\therefore \quad \lim_{n\to\infty} \sum_{k=1}^{\infty} S_k = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

41) $y.(y')^2 + xy' - yy' - x = 0$

$yy'[y' - 1] + x[y' - 1] = 0$

$$\frac{dy}{dx} = 1 \text{ or } \frac{dy}{dx} = -\frac{x}{y}$$

$y = x + c$ or $y^2 = -x^2 + c$

as curves passing through (3, 4)

$y = x + 1$

and $x^2 + y^2 = 25$

42) $\int \frac{f'(x) - f''(x)}{f(x) - f'(x)} dx = \int dx$ $f(x) - f'(x) = \lambda e^x$.

$e^{-x} f(x) = -\lambda x + c$

$f(x) = -\lambda x e^x + ce^x$

put $x = 0$ & $x = 1$ to get $c = 0$ & $\lambda = -3$.

$f(x) = 3x e^x$. ⇒ the equation $3x e^x = 0$ has only one solution i.e. $x = 0$.

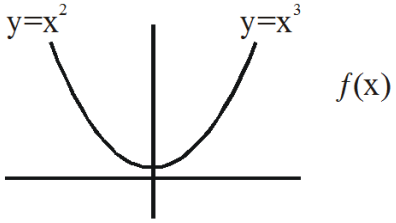
43) $\int_0^1 \left(\frac{e^x . x^3}{2} - e^x (3x) \right) dx$

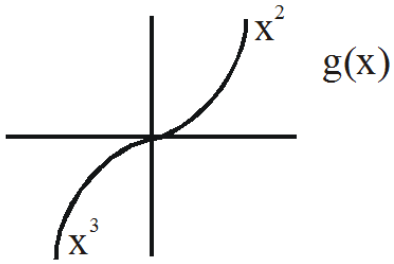
$= \int_0^1 \left(e^x \frac{x^3}{2} + e^x \frac{3x^2}{2} - e^x \frac{3x^2}{2} - e^x (3x) \right) dx$

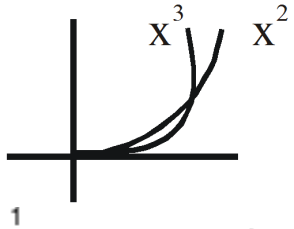
$= \int_0^1 e^x \left(\frac{x^3}{2} + \frac{3x^2}{2} \right) dx - \int_0^1 e^x \left(\frac{3x^2}{2} + 3x \right) dx$

$= \left(e^x . \frac{x^3}{2} \right)\Big|_0^1 - \left(e^x \frac{3x^2}{2} \right)\Big|_0^1 \Rightarrow \frac{e}{2} - \frac{3e}{2} = -e$.

44)

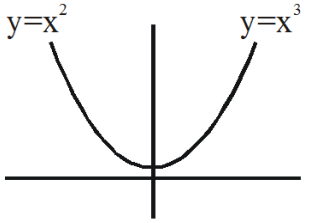




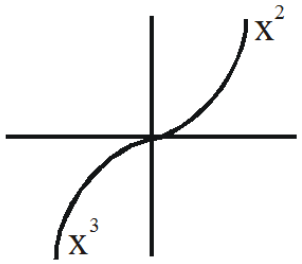


$\int_0^1 \left(x^2 - x^3 \right) dx = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

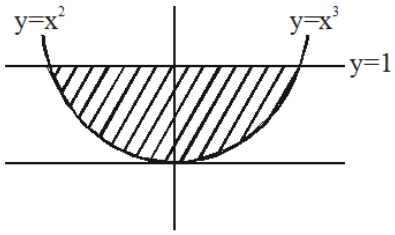
45)



$f(x)$



$g(x)$



$$2-\left(\int_0^1 x^3dx+\int_{-1}^0 x^2dx\right)$$

$$2-\left(\frac{1}{4}+\frac{1}{3}\right)=2-\frac{7}{12}=\frac{17}{12}$$

46) $f'(x) = 6x^2 + 2bx + 4 + \cos^2 x$

$f'(x) > 0 \quad \forall x \in R$

$6x^2 + 2bx + 4 + \cos^2 x + > 0$

$6x^2 + 2bx + 4 > 0$

$3x^2 + bx + 2 > 0$

$D < 0,\ b^2 - 24 < 0$

$-\sqrt{24} < b < \sqrt{24}$

$b = 0, \pm1, \pm2, \pm3, \pm4$

47)
$$k=\int_{-2013}^{2013}\frac{1}{\left(1+x^{2011}+\sqrt{1+x^{4022}}\right)}\frac{\left(1+x^{2011}-\sqrt{1+x^{4022}}\right)}{\left(1+x^{2011}-\sqrt{1+x^{4022}}\right)}.dx$$

$$=\int_{-2013}^{2013}\frac{1+x^{2011}-\sqrt{1+x^{4022}}}{2.x^{2011}}.dx$$

$$=\int_{-2013}^{2013}\underbrace{\frac{1}{2.x^{2011}}}_{\text{odd function}}dx+\int_{-2013}^{2013}\frac{1}{2}.dx-\int_{-2013}^{2013}\underbrace{\frac{\sqrt{1+x^{4022}}}{x^{2011}}}_{\text{odd function}}dx$$

$$=\int_{-2013}^{2013}\underbrace{\frac{1}{2.x^{2011}}}_{\text{odd function}}dx+\int_{-2013}^{2013}\frac{1}{2}.dx-\int_{-2013}^{2013}\underbrace{\frac{\sqrt{1+x^{4022}}}{x^{2011}}}_{\text{odd function}}dx$$

48) Divide given diff. equation by xy $(y^2 - x^2)$.

$$\frac{y(y^2-2x^2)}{xy(y^2-x^2)}dx + \frac{x(2y^2-x^2)}{xy(y^2-x^2)}dy = 0$$

$$\Rightarrow \frac{y^2-x^2-x^2}{x(y^2-x^2)}dx + \frac{y^2+y^2-x^2}{y(y^2-x^2)}dy = 0$$

$$\Rightarrow \frac{dx}{x} - \frac{x\cdot dx}{y^2-x^2} + \frac{ydy}{y^2-x^2} + \frac{dy}{y} = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} + \frac{ydy - xdx}{y^2-x^2} = 0$$

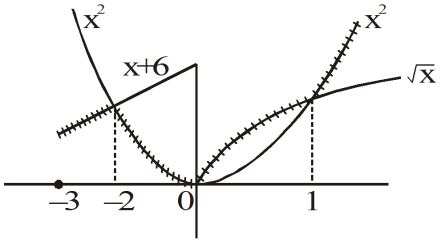
$$\Rightarrow \ln x + \ln y + \frac{1}{2}\ln(y^2-x^2) = k$$

$$\Rightarrow x^2\cdot y^2(y^2-x^2) = c$$

$$\Rightarrow f(x) = x^2,\ \ g(y) = y^2$$

49) $f : [-3, 1] \to R$

$$f(x) = \begin{cases} \min\left\{(x+6),\ x^2\right\} & , \quad -3 \leqslant x \leqslant 0 \\ \max\left\{\sqrt{x},\ x^2\right\} & , \quad 0 \leqslant x \leqslant 1 \end{cases}$$



area bounded by y = f(x) and x-axis

$$= \int_{-3}^{-2}(x+6)dx + \int_{-2}^{0}x^2dx + \int_{0}^{1}\sqrt{x}\,dx$$

$$A = \frac{41}{6}$$

$6A = 41$

50) $2\sqrt{2}\cos 15^\circ = 4\sin 45^\circ \cos 15^\circ$

$= 2\ (\sin 60^\circ + \sin 30^\circ)$

$$= \left(\sqrt{3}+1\right)$$

Now $$\int \frac{2\cos x\,dx}{\left((\sin x+1)^{1010}(1-\sin x)^{3030}\right)^{\frac{1}{2020}}}$$

$$= \int \frac{2\cos x\,dx}{(1+\sin x)^2\left(\frac{1-\sin x}{1+\sin x}\right)^{3/2}}$$

Put $\frac{1-\sin x}{1+\sin x} = t$

$$\frac{(-\cos x\,(1+\sin x) - (1-\sin x)\cos x)}{(1+\sin x)^2}dx = dt$$

$$\Rightarrow \frac{2\cos x\,dx}{(1+\sin x)^2} = -dt$$

$= -\int \frac{dt}{t^{3/2}} = \frac{2}{\sqrt{t}} + c$

So, $\int_0^{\pi/6} \frac{\left(2\sqrt{2}\cos 15^\circ\right)\cos x\,dx}{\left((\sin x+1)^{1010}(1-\sin x)^{3030}\right)^{\frac{1}{2020}}}$

$= \left(1+\sqrt{3}\right)\left(\sqrt{3}-1\right) = 2$

51) $p = \lim_{n\to\infty}\left(\left(\frac{1+n}{n}\right)\left(\frac{2+n}{n}\right)\left(\frac{3+n}{n}\right)\ldots\left(\frac{n+n}{n}\right)\right)^{\frac{1}{n}}$

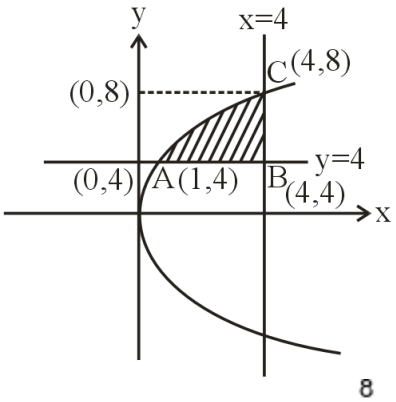
$\ln p = \lim_{n\to\infty}\frac{1}{n}\left(\sum_{r=1}^{n}\ell n\left(1+\frac{r}{n}\right)\right)$

$= \int_0^1 \ell n\,(1+x)\,dx$

$\ln p = \frac{4}{e}$

$\Rightarrow p = \frac{4}{e} \Rightarrow pe = 4$

$\Rightarrow x = 4$ & $y = 4$ & $y^2 = 16x$



Required Area (A) = $16 - \int_4^8 \frac{y^2}{16} = \frac{20}{3}$

$\Rightarrow [A] = 6$